Abstract:

Geographic barriers to health care remain a major determinant of inequity in Peru, disproportionately affecting rural and Amazonian populations. The country's complex topography and fragmented transport networks create substantial delays in reaching health services, threatening the achievement of Universal Health Coverage (UHC). Estimating travel time to the nearest facility provides a critical lens to identify spatial inequalities in effective health access. We developed a national geospatial model of physical accessibility, estimating the minimum travel time from any point in Peru to the closest health facility. The analysis integrated open-source data on health facility locations (RENIPRESS 2021), road networks (OpenStreetMap), land cover (Copernicus 2019), and elevation (SRTM, 30 m), combined with administrative boundaries from INEI. Travel speeds were parameterized by road class, slope, and land cover, applying a least-cost path algorithm (Dijkstra) implemented via the *gdistance* package in R. Marked inequalities were observed across ecological regions. Over 80% of coastal and inter-Andean residents can reach a facility within 30 minutes, whereas many Amazonian communities require more than 6 hours and, in extreme cases, over 15 hours. Regression analyses identified the State Density Index (adjusted PR 0·046, 95% CI 0·01-0·21; $p < 0·001$) and population size (adjusted PR 0·993, 95% CI 0·985-0·999; $p < 0·05$) as independent predictors of travel time to Category III facilities, highlighting the role of governance and demographic concentration in shaping access. Profound territorial inequities persist in physical access to health care in Peru, reflecting historical patterns of exclusion and infrastructure deficits. These findings underscore the urgency of targeted investments in road connectivity, telehealth, and strategic deployment of higher-complexity services to close geographic gaps and advance UHC.

Introduction:

Geographic accessibility is a major factor of health equity and a core requirement for achieving Universal Health Coverage (UHC).[1] In many low- and middle-income countries (LMICs), including those in Latin America, long distances and travel time to health facilities still delay or prevent care, disproportionately affecting rural and indigenous populations.[2-4] Peru is a clear example of this situation. Although insurance coverage has expanded in recent years, important structural gaps remain in where services are located and how easily people can reach them, particularly in Andean and Amazonian areas.[5,6] Evidence indicates that physical access strongly influences health outcomes such as skilled birth attendance, child survival, and chronic disease management.[7-9] For this reason, geographic accessibility increasingly used as an equity indicator in global health monitoring frameworks, including those linked to the Sustainable Development Goals.[10]

Yet, the spatial dimension of health inequity in Peru remains poorly quantified at a national scale. While regional studies have assessed access barriers in isolated Amazonian communities or specific rural provinces, few analyses have integrated high-resolution geospatial modeling with facility categorization to estimate travel times across the entire Peruvian territory.[11] Existing evidence often lacks granularity, overlooks facility capacity differences (Levels I-III), and fails to systematically identify populations facing extreme delays.[12-14] Furthermore, although geospatial approaches such as cost-distance modeling and least-cost path algorithms have been widely applied in Africa and Asia to inform service planning,[15,16] their use in the Peruvian

context has been limited, leaving critical gaps in understanding territorial inequities in health access.[17,18]

This study addresses these gaps by developing a national geospatial model to estimate minimum travel time from any point in Peru to the nearest health facility, categorized by level of care. We use open-source datasets and cost-distance modelling to address three main goals. First, we map how travel time to health facilities is distributed across the territory. Second, we estimate how many people face long travel times, disaggregating results by ecological region and by urban versus rural areas. Third, we reflect on what these patterns imply for fair service delivery and for progress toward universal health coverage. By providing a comprehensive, data-driven assessment of geographic barriers, this study offers actionable evidence to guide policy interventions targeting underserved populations and to inform national health system planning.

Methods:

Ethics Statement

This study relied exclusively on secondary, publicly available datasets, including health facility registries (RENIPRESS), demographic information from the National Institute of Statistics and Informatics (INEI), and geospatial layers from open-access sources such as OpenStreetMap, Copernicus Land Cover, and the Shuttle Radar Topography Mission (SRTM). All data were aggregated, anonymized, and published under open licenses (e.g., CC BY 4.0) without any personal identifiers. In accordance with national and international guidelines for the ethical use of secondary data, and following good research practices, this analysis did not require direct interaction with human

participants or access to identifiable information. Therefore, the study was exempted from review by the Institutional Review Board of the Universidad Señor de Sipán, as confirmed under category of research using publicly available data.

Study Setting

Peru (approximate centroid: 9·2°S, 75·2°W) is an upper-middle-income country in South America with an estimated population of 34 million people in 2024. The national territory spans 1,285,215 km² and exhibits marked ecological heterogeneity, comprising three major natural regions: the arid coastal plain (11·6% of the territory), the Andean highlands (28·1%), and the Amazon basin (60·3%). Peru has a highly diverse topography, which leads to strong variation in population density, transport infrastructure, and physical access to health services. Elevation ranges from sea level along the Pacific coast to around 6,768 m at the Andean peaks. This gradient creates major logistical challenges for delivering healthcare across the 24 departments and 196 provinces of the country. According to the Ministry of Foreign Affairs and European Union Cooperation (2024), these geographic conditions are closely linked to marked contrasts in transport networks and effective access to health care.

The health system itself is also fragmented. It brings together several public and private subsystems, including the Ministry of Health (MINSA), EsSalud, health services for the Armed Forces and the Police, and a wide range of private providers. In this study, we considered 34,214 health facilities registered in the National Registry of Health Service Provider Institutions (RENIPRESS) under SUSALUD.

According to national standards, these facilities are classified into three levels of care. Level I corresponds to primary care. Level II includes specialized care. Level III provides

high-complexity services. Their distribution is highly unequal. The coastal region, with major cities such as Lima and Trujillo, concentrates most level III hospitals and specialized units, which reflects the urban centralization of advanced services.

In contrast, the Amazon region covers more than half of the national territory but is home to only about 13% of the population and less than 8% of level III facilities. These establishments are mostly located in departmental capitals such as Iquitos. As a result, many rural and riverine communities depend almost entirely on low-complexity level I facilities. This situation reinforces existing geographic and structural inequities in timely and high-quality access to health care.

Study design and population

In this study, we carried out an ecological, cross-sectional analysis to examine how travel time to health facilities is distributed across Peru. Our focus was on physical accessibility to health establishments listed in the *Registro Nacional de Instituciones Prestadoras de Servicios de Salud* (RENIPRESS). These facilities are classified by the Ministry of Health into three levels of complexity: Level I for primary care, Level II for specialized care, and Level III for high-complexity services. The study population comprised the entire Peruvian population (~34 million inhabitants in 2024), as represented by gridded population estimates derived from the most recent national census and complementary demographic projections. Because the primary objective was to quantify spatial accessibility rather than individual-level characteristics, population data were integrated at a high spatial resolution and linked to modeled

travel times to the nearest health facility. No individual identifiers were used at any stage of the analysis.

Data Sources and Preparation

We integrated five open-access datasets to construct a national geospatial model of healthcare accessibility in Peru. The core input was the *Registro Nacional de Instituciones Prestadoras de Servicios de Salud* (RENIPRESS, SUSALUD, 2021), which lists 34,214 facilities categorized by complexity levels I (primary care), II (specialized care), and III (high-complexity hospitals), based on Technical Health Standard No. 021-MINSA/DGSP-V.03. Territorial and environmental data included: (1) the national road network from OpenStreetMap (2025), classified by functional hierarchy; (2) Copernicus Global Land Cover (2019), differentiating urban areas, agricultural land, forest, water bodies, and snow; (3) a Shuttle Radar Topography Mission (SRTM) digital elevation model (90 m resolution) for slope correction; and (4) official administrative boundaries from INEI and IGN. We reprojected all spatial layers to a common coordinate system, UTM Zone 18S (EPSG: 32718), so that all distance measurements were in consistent metric units.

Friction Surface Construction

We built the friction surface for the accessibility model by estimating, for each 1 km^2 raster cell, the number of minutes required to cross it. This estimate combined information on the road network, terrain slope, and land cover. Road vectors were converted to a raster format, and each cell intersected by a road was assigned the highest applicable base speed according to road class: 100 km/h for motorways and

trunk roads, 80 km/h for primary roads, 60 km/h for secondary roads, 50 km/h for tertiary roads, and 35 km/h for residential or unclassified roads. To reflect topographic variation, we then adjusted these base speeds using Tobler's hiking function[19] for pedestrian travel and a penalty factor for motorized transport:

$W_{i,t} = 6e^{-3\cdot5|\tan\theta_{i,t}+0\cdot05|}$, where:

$W_{i,t}$ represents the speed of hiker I (in km/h), at time t, while $\tan\theta_{i,t}$ represents the slope calculated as elevation difference/distance (dh/dx). In areas lacking vehicular access, pedestrian travel times were further modified by land cover-specific friction coefficients: urban areas (1·0), agricultural land (1·2), rocky terrain (1·8), forest and wetlands (2·5-3·0), and water or snow surfaces (6·0). The final friction surface was computed by integrating these components multiplicatively as applied to Peru by Carrasco-Escobar et al[3], with missing values replaced by neutral coefficients to maintain consistency across the modeled surface.

Cost-Distance Analysis

Travel time to the nearest health facility was calculated using a least-cost path algorithm[20] implemented through the *gdistance* package in R (version 2024·03).[21] The process involved: i) Generating a transition matrix representing movement cost between adjacent cells; and, ii) computing cumulative travel time (minutes) from each cell to the nearest health facility using the acCost function. The output was a high-resolution travel time mosaic for all facilities and stratified by category (I, II, III).

Facility Categorization and Access Thresholds

Health facilities were grouped into three complexity levels: Level I (basic and primary care), Level II (specialized hospitals), and Level III (high-complexity hospitals and national institutes). Sublevels were grouped rather than analyzed separately in order to keep the analysis clear and interpretable. To read the results from an equity perspective, we classified travel times into four strategic categories, following WHO and Lancet Commission recommendations for emergency and surgical care: up to 1 hour (essential access), 1 to 2 hours (considered adequate for emergency response), 2 to 4 hours (limited access with higher clinical risk), and more than 4 hours (a critical gap in access). These thresholds reflect evidence that prolonged travel delays significantly increase mortality in obstetric emergencies, trauma, and sepsis.[22,23]

Population Integration and Equity Analysis

Population distribution was integrated using gridded demographic estimates from INEI and WorldPop[24] to calculate the proportion of inhabitants within each access threshold. Results were disaggregated by administrative region (24 departments + Callao), ecological zone (coast, highlands, Amazon), and urban-rural classification. Maps and summary statistics were generated in QGIS 3·34 and R, ensuring reproducibility under STROBE and RECORD reporting guidelines.

Results:

National Overview of Travel Time

Travel time to health facilities in Peru varied substantially across administrative regions and increased markedly with the complexity of care. For category I facilities (primary care), the median travel time nationwide was 2·3 hours, compared with 9·1 hours for category II and over 30 hours for category III. Accessibility improved in densely populated coastal areas and declined sharply in the Andean highlands and Amazon basin.

Department-Level Disparities

We summarized the average and median travel times (in minutes) to facilities by category and department (Table 1). For category I facilities, the departments with the longest mean travel times were Madre de Dios (874·2 min), Ucayali (677·5 min), and Loreto (609·5 min). In contrast, Callao averaged less than 5 minutes, and Lima and Lambayeque stayed below 3 hours (Fig. 1A). Median travel times showed a similar pattern, with the highest values in Madre de Dios (728·3 min) and Ucayali (545 min).

For category II, the longest mean times were observed in Loreto (3212·4 min), Ucayali (2929·1 min), and Madre de Dios (2267·7 min), while Callao again showed almost no delay (3·9 min) and Lima remained relatively low (358·1 min) (Fig. 1B).

The largest gaps appeared in category III. Amazonas (6599·5 min), Loreto (5038·1 min), and Ucayali (3383·8 min) faced extremely long travel times, whereas Callao had the shortest access (8·8 min), followed by Lambayeque (684·1 min) and Lima (437·8 min) (Fig. 1C).

Spatial Patterns and Critical Zones

At the national level, travel times to category I, II, and III facilities showed wide variation (Fig. 2). Almost all Amazonian and high-Andean provinces exceeded the 4 hour threshold recommended for emergency and surgical care, and many areas surpassed 24 hours (Table 2). Compared with category III facilities (Fig. 2C), travel times improved moderately, but large disparities remained for category I (Fig. 2A) and category II facilities (Fig. 2B). Coastal and some Andean departments (Callao, Lima, Ica, Moquegua, Junín, Puno) had median travel times below 4 hours, whereas Loreto, Ucayali, Madre de Dios and Amazonas exceeded 30 hours, with extreme values over 160 hours in remote settlements. Category I facilities showed denser territorial coverage, and most coastal and Andean departments reported medians under 4 hours (Fig. 1A). Even so, Amazonian departments continued to experience very long travel times, with interquartile ranges extending beyond 60 hours for highly isolated rural and riverine communities and for most provinces (Supplementary File 1).

Equity Perspective

Across all facility categories, our results show a clear gradient of geographic inequity that follows the country's main ecological regions. The Amazon carries the heaviest burden of delayed access, whereas large urban areas, especially Lima and Callao, have almost immediate access to higher-level care. These contrasts reveal a pattern of structural territorial exclusion. This exclusion is already visible at the primary care level and becomes even more pronounced for specialized services.

The multivariate models show that these patterns are not random. Instead, they are systematically linked to state presence and to where people are concentrated (Table 3). The State Density Index (IDE), used as a proxy for territorial state capacity, was

significantly associated with shorter travel times to level II and level III facilities. For access to level III care, the IDE coefficient was 0.046 (95% CI: 0.01 to 0.21, $p < 0.01$). This result suggests that stronger state presence is associated with better access to specialized care, independent of population size.

Population size was also inversely and significantly associated with travel time across all levels of care, particularly for level I and level III services. For level III access, the population coefficient was 0.993 (95% CI: 0.985 to 0.999, $p < 0.05$). Taken together, these findings support the idea that both state infrastructure density and population concentration act as protective factors against geographic exclusion. They also underline the urgent need to expand health services in territories with low state presence and dispersed populations.

Discussion:

This study shows deep and persistent territorial gaps in physical access to healthcare across Peru. Using a high-resolution national geospatial model, we estimated travel times to health facilities of all complexity levels and found a clear pattern. Access to primary care (level I) is generally better than access to specialized care, yet critical gaps remain in the Amazon and in high-Andean areas. Travel times rise sharply as facility complexity increases. More than 80% of the country can reach a level I facility within three hours, but only a small share of the population can get to level II or III services in under four hours. In departments such as Loreto, Ucayali, Amazonas, and Madre de Dios, people often need more than 24 hours of travel to reach a hospital. This situation falls well outside the WHO benchmark of ≤1 hour for essential care and the Lancet Commission's recommendation of ≤2 hours for emergency surgical services.[25,26] In

practice, this means that Peru's health system, even after achieving high levels of insurance coverage, still operates in a geographically inequitable way, especially in territories that are historically more vulnerable and structurally excluded.

Our results also add new detail to what previous studies had already highlighted. Earlier work by Carrasco-Escobar et al.[3] and Hernández-Vásquez et al.[4] provided important evidence on average accessibility for rural and indigenous populations in Peru. However, those analyses focused on primary care or on specific regions and did not differentiate by facility complexity or explore variation within departments. In contrast, our model shows that even in departments with moderate mean travel times, interquartile ranges can exceed 60 hours. This suggests that severe local inequities are hidden behind regional averages. These "last-mile" barriers fall most heavily on riverine, mountainous, and forest-based communities, echoing patterns described in sub-Saharan Africa and parts of Asia, where difficult terrain and historical underinvestment in health systems reinforce each other.[12,27,28] Our findings also question the usual idea of a simple urban–rural divide. Some peri-urban and highland populations experience delays that are far beyond recommended thresholds, which indicates that spatial health planning should be guided by effective travel time and functional proximity rather than only by administrative boundaries.

The regression models help to clarify which factors are driving these patterns. Higher state density was significantly associated with shorter travel times to level II and III services, even after adjusting for population size. For level III facilities, the IDE coefficient was 0·046 (95% CI: 0·01 to 0·21, $p < 0·01$). This suggests that stronger

territorial state presence is linked to better access to specialized care. Population size was also negatively associated with travel time across all facility levels. The effect was especially marked for level I and III services. For level III access, the population coefficient was 0·993 (95% CI: 0·985 to 0·999, $p < 0·05$). Taken together, these results support the idea that both state infrastructure density and demographic concentration act as protective factors against geographic exclusion. They also reinforce the view that inequities in health access are not random events but the result of structural patterns shaped by state capacity and the spatial distribution of settlements.

This study also has limitations. We did not explicitly model real-time disruptions, such as seasonal flooding, road damage, or changes in transport availability, which could make travel times even longer in remote areas. We also did not include information on facility readiness or quality of care, so we cannot fully assess effective coverage. At the same time, the study has important strengths. We relied on open and validated data sources, applied cost–distance algorithms that incorporate land cover, slope, and road hierarchy, and used nationwide coverage at a resolution of 1 km^2.

This approach increases both the internal and ecological validity of the estimates. Selection bias was limited by using a census-based gridded framework, while confounding was reduced by focusing on one clear spatial outcome. Information bias was addressed by harmonizing inputs from RENIPRESS, OpenStreetMap, and INEI and by using neutral coefficients where information was missing. The overall analytical strategy is consistent with STROBE and RECORD recommendations for spatial epidemiology.

Translating these findings into policy requires rapid and evidence-based decisions. First, Peru needs to expand and strengthen its primary healthcare network in rural and

indigenous territories, using an intercultural approach that involves community health workers, traditional healers, and culturally appropriate pathways of care.[2] Second, in departments such as Loreto, Ucayali, and Amazonas, where road connections are very limited, mobile health teams and river-based posts should be prioritized to guarantee essential care within ≤1 hour, in line with WHO and Lancet Commission standards.[25] Third, investing in satellite-supported telemedicine would allow remote level I and II facilities to provide specialized consultations, reduce unnecessary referrals, and improve continuity of care.[29] Fourth, multimodal infrastructure strategies are crucial. Investments need to focus on road and navigable river corridors that link provincial capitals to level II-1 and II-2 hospitals, and on rural airstrips that can support emergency evacuations from highly isolated areas.[16] These interventions will only be sustainable if they are built into decentralized territorial investment plans that use geospatial information and travel time metrics to guide infrastructure projects and health budgets. In addition, effective coverage needs to be monitored on an ongoing basis, not just physical access, to ensure that service expansion actually results in timely and equitable care.[30] When spatial accessibility is treated as a core health system indicator, Peru and other middle-income countries can move beyond nominal insurance coverage and begin to close the persistent geographic gaps that undermine Universal Health Coverage.[31]

Contributors:

ETO, CBL, FMD, and MRM conceptualized the study, designed the geospatial modeling framework, and led the literature review. AMQ supervised the project as part of the final evaluation for the Biostatistics for Public Health course at the Universidad de

Ingeniería y Tecnología (UTEC). ETO, CBL and MRM conducted data acquisition and preprocessing; ETO and CBL developed the friction surface and implemented the travel time models using R and QGIS. ETO, FMD, and MRM were responsible for map creation, figure design, and statistical summaries. AMQ provided methodological guidance, verified the underlying data, and led the development of equity metrics. ETO, CBL, FMD, and MRM wrote the first draft of the manuscript in Spanish and AMQ wrote the first draft in English. All authors contributed to the interpretation of findings, critically reviewed the manuscript for important intellectual content, approved the final version, and agreed to be accountable for the integrity and accuracy of the work.

Availability of data and materials:

The data supporting this study's findings are open data curated by the Global Health Data Science Initiative, which is available under the Creative Commons Attribution 4·0 (CC BY 4·0) license. The data and code are freely available in:

- Data: https://doi.org/10.5281/zenodo.17342580 (accessed on October 13, 2025)
- Code: https://github.com/EduardoTllo/Mapping-Inequity

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Figure legends:

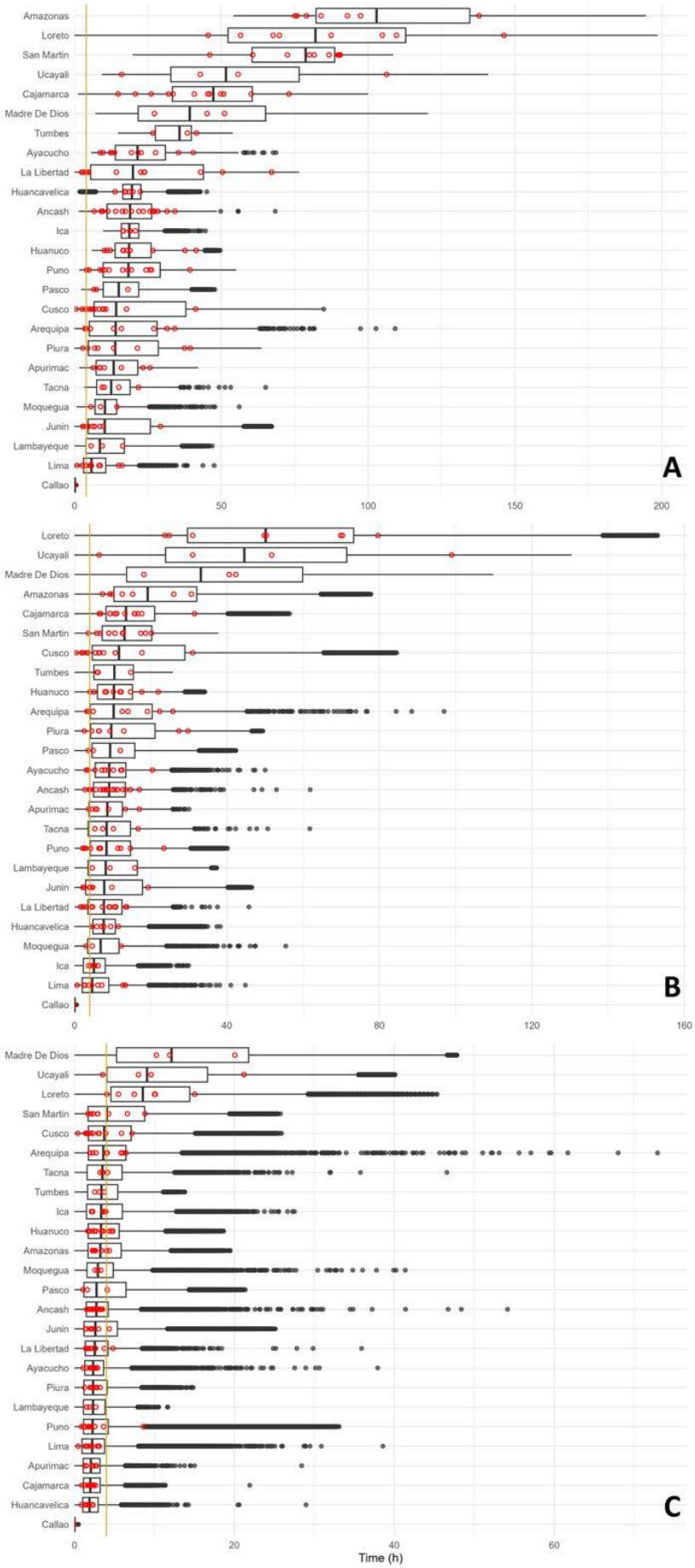
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Figure 1: Department-level distribution of travel time to nearest health facility by level of care, Peru, 2025

Legend: Boxplots show the distribution of estimated travel times (in minutes) from population centroids to the nearest health facility of level I (fig. 1A), level II (fig. 1B), and level III (fig. 1C) across Peru's 24 departments and the constitutional province of Callao. Red circles indicate departmental means, boxes show medians and interquartile ranges, and whiskers denote 1·5×IQR. Travel times increase markedly with facility complexity, reflecting widening inequities in access to higher levels of care. The most extreme delays are observed in Amazonian departments—particularly Loreto, Ucayali, Madre de Dios, and Amazonas—where median travel times often exceed 30 hours. In contrast, coastal and Andean departments typically maintain median travel times under 4 hours for level I facilities. The shortest times across all levels occur in Callao. These patterns highlight persistent geographical barriers to equitable access to secondary and tertiary healthcare services in Peru.

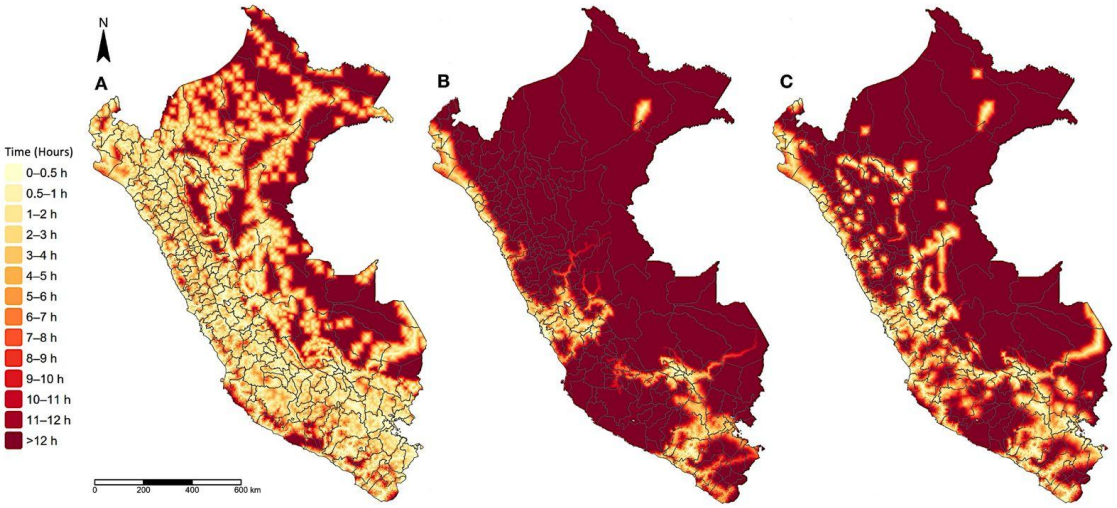

Figure 2: National distribution of travel time to nearest health facility by level of care, Peru, 2025

Legend: Maps show the estimated minimum travel time (in hours) to the nearest health facility across Peru, stratified by level of care: level III (fig. 2A), level II (fig. 2B), and level I (fig. 2C). Travel times were modeled using a cost-distance algorithm incorporating road networks, terrain slope, and land-cover data at 1 km² resolution. Accessibility declines sharply with increasing facility complexity: while most districts have level I facilities within 3 hours, large portions of the Amazon basin, the central Andes, and other remote areas require more than 12 hours to reach level II or III facilities. The best access to high-complexity care is concentrated in major urban and coastal regions, particularly Lima, Callao, and the northern coast. These spatial patterns reveal pronounced territorial inequities in physical access to essential and specialized healthcare services.

Table 1. Average and median accessibility time (minutes) to healthcare centers by region and category of establishment*.

	Category I		Category II		Category III	
	Mean	Median	Mean	Median	Mean	Median
Amazonas	6599·5	6174·0	1378·8	1153·6	252·9	194·2
Loreto	5038·1	4926·0	3212·4	3010·5	609·5	513·0
San Martin	4467·6	4724·5	842·5	786·7	356·8	243·3
Ucayali	3383·8	3096·5	2929·1	2674·3	677·5	545·0
Cajamarca	2829·8	2839·2	947·0	812·0	142·3	119·0
Madre De Dios	2682·2	2355·7	2267·7	1989·7	874·2	728·3
Tumbes	2032·1	2147·8	626·4	628·7	232·3	202·8
La Libertad	1529·8	1195·4	524·5	463·8	181·4	152·9
Cusco	1370·3	851·9	1143·1	701·4	301·3	223·3
Ayacucho	1367·6	1288·9	596·5	547·1	165·9	140·4
Huánuco	1277·2	1116·5	668·1	622·0	248·3	199·5
Puno	1238·3	1102·9	653·0	504·1	279·6	136·0
Huancavelica	1170·6	1175·0	508·8	459·5	130·5	113·2
Ica	1157·4	1121·8	337·6	307·3	252·6	200·8
Ancash	1135·9	1135·8	563·6	545·9	188·5	164·3
Piura	1086·1	833·1	808·0	580·2	177·7	139·8

Arequipa	1052·5	842·6	786·1	617·5	299·1	216·4
Junín	1006·3	620·5	682·5	469·3	238·0	156·9
Pasco	995·5	905·8	682·0	562·1	277·2	165·7
Apurimac	891·1	802·3	526·1	518·0	134·6	123·1
Tacna	804·7	751·7	554·4	506·7	252·3	210·1
Lambayeque	684·1	519·9	644·4	492·4	157·7	139·3
Moquegua	655·5	623·6	473·0	412·7	234·0	176·0
Lima	437·8	347·4	358·1	279·0	163·1	133·7
Callao	8·8	8·7	3·9	2·2	1·5	0·0

* Level of healthcare facility (Categories I-III): Category I corresponds to basic primary-level facilities, Category II to intermediate care centers, and Category III to hospitals or high-complexity institutions.

Table 2. Median travel times (hours) to category II and III healthcare facilities by region and province.

	Provinces	Median ± IQR (h)		< 1h		1-< 4 h		≥ 4h	
		II	III	II	III	II	III	II	III
		12·57 ± 11·09	84·28 ± 17·87	0	0	0	0	7	7
Ancash	20	8·45 ± 3·59	20·70 ± 13·92	0	0	2	0	18	20
Apurimac	7	5·94 ± 6·19	10·27 ± 11·07	0	0	1	0	6	7
Arequipa	8	13·22 ± 15·05	14·59 ± 22·80	0	0	2	1	6	7
Ayacucho	11	8·93 ± 5·98	19·42 ± 12·62	0	0	2	0	9	11
Cajamarca	13	11·08 ± 6·77	45·78 ± 18·42	0	0	0	0	13	13
Callao	1	0·04 ± 0·00	0·14 ± 0·00	1	1	0	0	0	0
Cusco	13	6·37 ± 4·45	6·37 ± 4·45	1	1	4	2	8	10
Huancavelica	7	7·29 ± 3·41	17·44 ± 1·96	0	0	0	0	7	7
Huánuco	11	10·42 ± 6·91	16·73 ± 8·48	0	0	0	0	11	11
Ica	5	5·05 ± 1·05	18·84 ± 2·23	0	0	1	0	4	5
Junín	9	4·53 ± 0·87	6·19 ± 4·32	0	0	3	2	6	7
La Libertad	12	8·21 ± 6·65	18·50 ± 24·27	0	0	3	2	9	10
Lambayeque	3	9·28 ± 5·52	9·48 ± 5·43	0	0	0	0	3	3
Lima	10	4·12 ± 4·04	5·57 ± 4·59	1	1	4	2	5	7
Loreto	8	50·16 ± 40·14	78·32 ± 40·68	0	0	0	0	8	8
Madre de Dios	3	40·61 ± 12·11	45·28 ± 11·95	0	0	0	0	3	3
Moquegua	3	4·63 ± 4·59	9·02 ± 4·44	0	0	1	0	2	3
Pasco	3	4·91 ± 4·23	7·48 ± 5·84	0	0	1	0	2	3
Piura	8	7·84 ± 10·73	10·65 ± 19·28	0	0	1	1	7	7
Puno	13	6·62 ± 7·15	16·31 ± 14·83	0	0	3	0	10	13
San Martin	10	11·62 ± 8·85	84·37 ± 15·52	0	0	1	0	9	10
Tacna	4	8·80 ± 5·03	12·76 ± 7·09	0	0	0	0	4	4
Tumbes	3	6·18 ± 4·41	38·38 ± 7·47	0	0	0	0	3	3
Ucayali	4	41·34 ± 38·70	49·24 ± 32·25	0	0	0	0	4	4

* Median and interquartile ranges (IQR) of travel times to Category II and III health facilities are shown for each region, along with the distribution of provinces by time-to-access strata

(< 1 h, 1 - 4 h, ≥ 4 h). Regions with higher proportions in the ≥ 4 h category denote critical barriers in physical accessibility to essential health services.

Table 3. Factors associated with travel time to category III healthcare facilities.

	Unadjusted PR	CI 95%	Adjusted PR	CI 95%
State Density Index (IDE)	0·036	0·012 to 0·106*	0·046	0·01 to 0·21**
Human Development Index (IDH)	0·021	0·004 to 0·122*	1·414	0·119 to 17·74
Population	0·985	0·978 to 0·992*	0·993	0·985 to 0·999*

Abbreviations:* PR = prevalence ratio; CI 95% = 95% confidence interval; *p < 0·05; **p < 0·001.

Supplementary Files:

Supplementary File 1: Supplementary Figures and Tables - Geographical Accessibility to Health Facilities in Peru, 2025. This appendix provides supporting figures and data for the analysis of spatial accessibility to health facilities in Peru. Travel times were estimated using a cost-distance algorithm that integrates road networks, terrain slope, and land-cover data at 1 km² resolution. All estimates refer to the modeled minimum travel time from population centroids to the nearest facility, by level of care.